

SMART RAILWAY WASHROOM AUTOMATION SYSTEM

A MINI PROJECT REPORT

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ABSTRACT

The “SMART RAILWAY WASHROOM AUTOMATION SYSTEM” is an IoT-based automated hygiene management system designed to improve sanitation in railway toilets. The system uses an ESP32 microcontroller along with PIR, MQ gas, and DHT11 sensors to monitor user activity, odour levels, and temperature conditions in real time. Based on sensor data, the system automatically controls flushing and ventilation mechanisms to maintain cleanliness and air quality. An LCD display provides real-time monitoring information and system status updates. The proposed system reduces manual intervention, minimizes water wastage, improves passenger hygiene experience, and provides a cost-effective and smart solution for railway sanitation automation.

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LIST OF ABBREVIATION

CPU	CENTRAL PROCESSING UNIT
DHT11	DIGITAL HUMIDITY AND TEMPERATURE SENSOR
ESP32	ESPRESSIF 32-BIT MICROCONTROLLER
IOT	INTERNET OF THINGS
LCD	LIQUID CRYSTAL DISPLAY
MQ	METAL OXIDE SEMICONDUCTOR GAS SENSOR
PIR	PASSIVE INFRARED SENSOR
SC	STRUCTURE CHART
UML	UNIFIED MODELLING LANGUAGE
WI-FI	WIRELESS FIDELITY

CHAPTER 1

INTRODUCTION

Cleanliness and hygiene in public transportation systems have become major concerns in recent years, especially in railway transportation where thousands of passengers travel daily. Among the various sanitation-related challenges faced by railways, maintaining the hygiene of train toilets remains one of the most difficult tasks. Traditional railway toilet maintenance systems mainly depend on manual inspection and fixed cleaning schedules, which are often inefficient and unable to respond to real-time environmental conditions. As a result, passengers frequently experience unpleasant odour, unhygienic conditions, water wastage, and delayed cleaning operations. These issues not only reduce passenger comfort but also create health and environmental concerns.

To address these problems, the project titled “Smart Railway Washroom Automation System” proposes an intelligent and automated solution for monitoring and maintaining train toilet hygiene using IoT and embedded system technologies. The system is designed to continuously monitor toilet conditions in real time and automatically perform necessary actions such as flushing and ventilation based on environmental conditions and user activity. The proposed solution aims to improve sanitation standards, reduce manual intervention, and ensure efficient resource management in railway toilets.

The system uses an ESP32 microcontroller as the core processing unit due to its high performance, low power consumption, and wireless communication capabilities. Multiple sensors are integrated into the system to monitor different parameters inside the toilet environment. A PIR (Passive Infrared) sensor is used to detect user presence and count usage frequency. An MQ gas sensor detects odour and harmful gas levels, while a DHT11 sensor measures temperature

conditions. These sensors continuously send data to the ESP32 controller, where rule-based AI logic analyzes the collected information and determines the appropriate actions. Based on the sensor readings, the system automatically activates a water pump to perform flushing operations and controls a ventilation fan to remove unpleasant odour and maintain proper air circulation. An LCD display is used to provide real-time information about system status, sensor readings, and operational conditions. Unlike camera-based monitoring systems, the proposed system ensures complete privacy by relying only on sensor-based detection mechanisms. One of the major advantages of this system is its ability to perform usage-based smart cleaning instead of following fixed cleaning schedules. This helps in reducing unnecessary water consumption and improves the efficiency of maintenance operations.

The system also operates independently without requiring continuous internet connectivity, making it highly suitable for railway environments where network availability may be limited or unstable during travel. Since the proposed system performs real-time monitoring and automated control operations locally using the ESP32 microcontroller, it can continue functioning efficiently even in offline conditions. This improves system reliability and ensures uninterrupted hygiene management throughout the journey.

The proposed project demonstrates how IoT, automation, and embedded intelligence can be effectively integrated to solve real-world sanitation problems in public transportation systems. By combining sensors, automated control mechanisms, and real-time environmental monitoring, the system improves hygiene maintenance while reducing manual intervention and operational complexity. The integration of PIR sensors, MQ gas sensors, DHT11 sensors, automated flushing systems, ventilation control, and LCD monitoring creates a

complete smart sanitation management solution suitable for railway applications.

Another important advantage of the proposed system is its low-cost and scalable architecture. The use of affordable embedded components and sensor modules makes the system economically feasible for large-scale deployment across multiple railway coaches and public sanitation facilities. The modular design also allows easy maintenance, future upgrades, and integration with additional smart technologies.

In the future, the system can be enhanced with advanced Artificial Intelligence and Machine Learning algorithms for predictive cleaning and intelligent maintenance scheduling based on passenger usage patterns. Mobile application integration can provide real-time alerts and remote monitoring support for maintenance staff and railway authorities. Cloud connectivity and centralized railway management systems can also be integrated to monitor multiple sanitation units simultaneously from a single control center.

Additional features such as automated disinfectant spraying, water level monitoring, energy-efficient operation, fault detection systems, and smart analytics can further improve system performance and operational efficiency. Thus, the proposed system provides a smart, reliable, cost-effective, and scalable solution for improving hygiene standards, resource management, and passenger comfort in modern railway transportation systems.

1.1 MOTIVATION FOR WORK

Railway transportation is one of the most widely used public transportation systems in the world, carrying millions of passengers every day. Cleanliness and sanitation in railway facilities play a major role in ensuring passenger comfort, safety, and public health. Among all facilities available in trains, toilets are one of the most frequently used and difficult areas to maintain due to continuous passenger usage, poor ventilation, water leakage, odor accumulation, and limited maintenance time. Maintaining hygiene in railway toilets has therefore become a serious challenge for railway authorities.

In traditional railway systems, toilet cleaning and maintenance are mostly performed manually based on fixed schedules rather than actual usage conditions. Such methods are often inefficient because the hygiene condition of the toilet changes continuously depending on passenger traffic and environmental conditions. At times, toilets remain unclean for long durations before maintenance personnel arrive for inspection. This creates discomfort for passengers and may also lead to health-related problems due to bacterial growth and poor sanitation.

The growing need for cleaner and smarter public transportation systems motivated the development of the proposed project. The project focuses on improving sanitation standards in railway toilets using IoT and embedded automation technologies. The system continuously monitors environmental conditions and automatically performs hygiene control operations whenever required.

One of the major motivations behind this work is the need to reduce manual monitoring and maintenance efforts. Railway staff cannot continuously inspect every toilet compartment manually, especially in long-distance trains and crowded railway routes. This often results in delayed cleaning and poor

maintenance quality. By implementing real-time sensor-based monitoring, the proposed system can automatically identify hygiene-related issues and perform corrective actions without human intervention.

The project also addresses the issue of unpleasant odor inside railway toilets. Poor ventilation and continuous usage often create uncomfortable conditions for passengers. Odor accumulation not only affects passenger comfort but also creates an unhygienic environment. The integration of gas sensors and automatic ventilation systems helps maintain fresh air circulation and improve restroom conditions.

Another important motivation is the reduction of unnecessary water wastage. Traditional flushing systems depend entirely on manual operation, and users may either forget to flush properly or use excessive water. Since water management is very important in railway systems, automated flushing mechanisms can help reduce wastage by activating only when required.

The major motivations behind the proposed project include:

- To improve hygiene and sanitation standards in railway toilets.
- To provide real-time monitoring of restroom conditions.
- To reduce unpleasant odour using automatic ventilation systems.
- To minimize manual cleaning and monitoring efforts.
- To reduce unnecessary water wastage through automated flushing.
- To improve passenger comfort and travel experience.
- To implement smart automation using IoT and embedded systems.

- To create a low-cost and scalable hygiene management solution.
- To support smart railway and smart transportation infrastructure.
- To ensure privacy-preserving monitoring without using cameras.

The advancement of IoT and embedded system technologies is another strong motivation for this project. Modern IoT systems enable real-time monitoring, automation, wireless communication, and intelligent decision-making at low cost. Embedded controllers such as the ESP32 provide powerful processing capabilities while consuming minimal power. These technologies make it possible to design compact and efficient smart hygiene systems suitable for railway environments.

The ESP32 microcontroller used in the proposed system acts as the central processing unit that receives sensor data, processes environmental conditions, and controls automated output devices. The integration of multiple sensors such as PIR sensors, MQ gas sensors, and DHT11 temperature sensors demonstrates how real-time environmental monitoring can be achieved effectively using embedded technologies.

The PIR sensor detects user presence and usage frequency inside the toilet. The MQ gas sensor monitors odor levels and harmful gases, while the DHT11 sensor measures environmental temperature conditions. By combining these sensors with intelligent control logic, the system can automatically activate flushing systems and ventilation mechanisms whenever required.

Another major motivation for this project is the increasing importance of public health and sanitation awareness. Clean public sanitation facilities are essential for preventing infections and maintaining healthier environments. In public transportation systems where thousands of passengers use the same facilities

daily, maintaining hygiene becomes even more critical. The proposed automation system helps improve cleanliness standards and reduces direct contact with restroom surfaces, thereby supporting better public hygiene practices.

The project is also motivated by the concept of smart public infrastructure and smart transportation systems. Modern railway networks are gradually adopting automation and intelligent monitoring technologies to improve operational efficiency and passenger satisfaction. However, sanitation management systems still lack sufficient automation in many areas. The proposed system contributes towards the development of intelligent railway infrastructure by integrating automation, real-time monitoring, and IoT technologies into sanitation management.

The proposed system also provides several operational and environmental advantages such as:

- Efficient resource utilization.
- Automated environmental monitoring.
- Reduced operational costs.
- Faster maintenance response.
- Improved energy efficiency.
- Continuous system monitoring.
- Enhanced passenger satisfaction.
- Reduced dependency on manual labor.

- Better air quality management.
- Sustainable sanitation practices.

Another important motivation behind this project is privacy preservation. Some advanced monitoring systems use camera-based surveillance for occupancy detection and monitoring. However, such systems create serious privacy concerns, especially in restroom environments. The proposed system avoids camera-based monitoring entirely and instead uses non-visual sensor technologies to maintain privacy while still ensuring effective automation and monitoring.

The project also serves as an educational opportunity for applying theoretical concepts related to IoT, embedded systems, automation, and sensor integration in solving real-world engineering problems. It demonstrates how multiple hardware and software components can work together to create intelligent automation systems for public applications.

Future scalability is another motivating factor for the project. The proposed system can be further enhanced by integrating cloud computing, mobile applications, AI-based predictive maintenance, and centralized monitoring systems. Such improvements can enable large-scale implementation across railway stations, train compartments, and other public sanitation facilities.

The project also supports environmental sustainability by reducing unnecessary water usage and optimizing energy consumption. Automated operation ensures that resources are used only when required, thereby improving efficiency and reducing wastage.

In addition, the system contributes towards improving the overall travel experience of railway passengers. Clean and hygienic sanitation facilities significantly improve passenger satisfaction and create a more comfortable travel

environment. Better hygiene management also improves the public image and service quality of railway transportation systems.

Overall, the motivation for developing the system arises from the need to create a smarter, cleaner, safer, and more efficient railway sanitation management solution. By integrating IoT technologies, embedded systems, sensors, and automated control mechanisms, the project aims to overcome the limitations of traditional sanitation systems and contribute towards the development of intelligent public infrastructure. The proposed system demonstrates how automation and real-time monitoring technologies can improve hygiene standards, resource management, operational efficiency, and passenger comfort in modern railway transportation systems.

1.2 PROBLEM STATEMENT

In railway transportation systems, maintaining hygiene and cleanliness in train toilets is a major challenge that directly affects passenger health, comfort, and travel experience. Although cleaning and maintenance activities are regularly performed, most railway toilets still rely on fixed cleaning schedules and manual inspection methods rather than real-time environmental monitoring. This often leads to delayed cleaning operations, unpleasant odour, poor ventilation, excessive water wastage, and unhygienic conditions inside the toilet environment. Therefore, there is a need for an intelligent and automated system that can continuously monitor hygiene conditions and automatically perform necessary control actions based on real-time data.

The proposed Smart Railway Washroom Automation System aims to address these issues using IoT sensors, embedded systems, and automated control mechanisms. The system is designed to monitor toilet usage, odour levels, and

temperature conditions continuously and activate flushing and ventilation systems whenever required.

Complex Environmental Conditions: Train toilet hygiene depends on multiple environmental factors such as odour concentration, temperature variation, user frequency, and ventilation conditions. The challenge is to accurately collect and analyze data from multiple sensors to determine the actual hygiene status of the toilet.

Real-Time Monitoring Requirement: Existing railway toilet systems do not provide continuous monitoring and instant response mechanisms. The proposed system must support real-time data collection, processing, and automated action control for effective hygiene management.

Data Variability: Passenger usage patterns vary depending on travel routes, time, and passenger density. The system must be capable of adapting to changing environmental conditions while maintaining reliable performance and accuracy.

Efficient Automation and Control: The system should automatically activate flushing and ventilation systems only when necessary to avoid unnecessary water and power consumption while ensuring proper sanitation.

Privacy and Low-Cost Implementation: The solution must maintain passenger privacy by avoiding camera-based monitoring systems and should also remain cost-effective for large-scale railway deployment.

Thus, the proposed system focuses on creating a smart, automated, and efficient railway toilet hygiene management solution that improves cleanliness, reduces manual effort, and enhances passenger comfort.

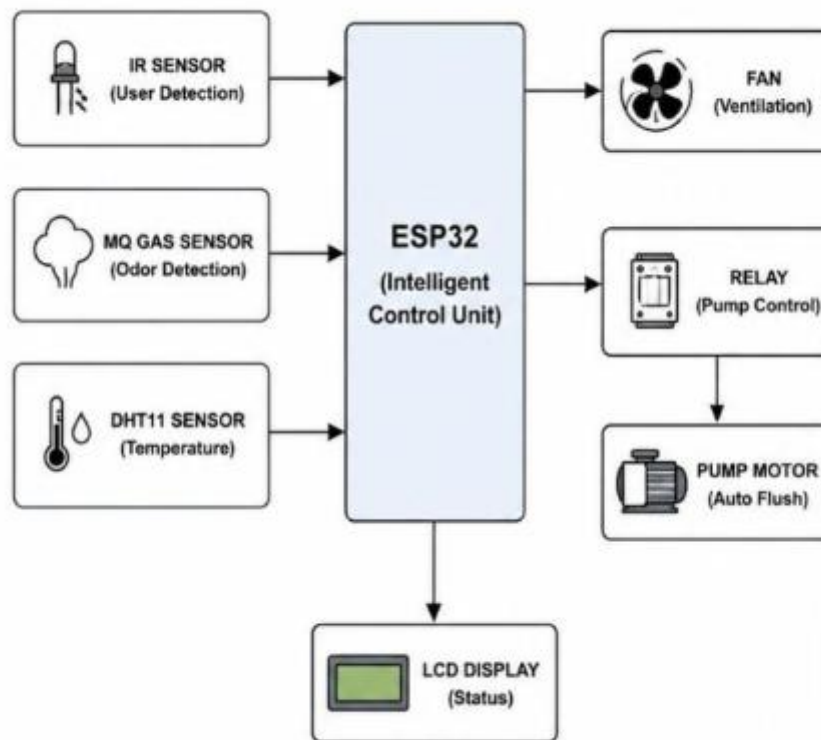


FIG 1.2.1 BLOCK DIAGRAM

The provided block diagram illustrates an **IoT-based Intelligent Automation System**, likely designed for a smart restroom or a controlled environment requiring hygiene and climate management. At its core, the system utilizes an **ESP32 microcontroller** as the brain to process environmental data and trigger mechanical responses.

Here is a breakdown of how the system works across its three primary stages: **Sensing, Processing, and Actuating**.

1. Input Stage: Environmental Sensing

The ESP32 continuously monitors three distinct types of data via specialized sensors:

- **IR Sensor (User Detection):** This sensor acts as the primary trigger for presence-based logic. It uses infrared light to detect when a person enters or leaves the vicinity. This ensures that resources like water or ventilation are only used when necessary, or immediately following use.
- **MQ Gas Sensor (Odor Detection):** This sensor monitors air quality by detecting specific gases (such as ammonia or sulfides). If odor levels exceed a pre-set threshold, it sends a signal to the ESP32 to initiate air purification or ventilation.
- **DHT11 Sensor (Temperature):** This digital sensor tracks ambient temperature and humidity. It allows the system to monitor the comfort level of the environment, ensuring it doesn't become too hot or humid, which could lead to bacterial growth or discomfort.

2. Processing Stage: The ESP32 "Brain"

The **ESP32 (Intelligent Control Unit)** serves as the central hub. It receives the analog or digital signals from the sensors and executes programmed logic. For example:

- **Logical Condition:** If IR Sensor = No User AND Timer > 5 seconds, then Trigger Auto Flush.
- **Logical Condition:** If Gas Level > Threshold, then Turn on Fan.

Because the ESP32 has integrated Wi-Fi and Bluetooth capabilities, this system could also theoretically send data to a cloud dashboard or a mobile app for remote monitoring, though the diagram focuses on local automation.

3. Output Stage: Actuation and Feedback

Based on the processing logic, the ESP32 controls three main output components:

- **Fan (Ventilation):** Managed by the MQ Gas sensor and DHT11, the fan is activated to exhaust odors or hot air. This maintains air quality and a neutral temperature within the space.
- **Relay & Pump Motor (Auto Flush):** Since the ESP32 cannot directly power a heavy-duty motor, it sends a low-voltage signal to a **Relay**. The relay acts as an electromagnetic switch, allowing a higher voltage circuit to power the **Pump Motor**. This motor drives the "Auto Flush" mechanism, ensuring hygiene without requiring manual contact.
- **LCD Display (Status):** This provides a human-machine interface. It displays real-time data such as current temperature, gas levels, or system status (e.g., "Flushing" or "System Ready"). This is vital for maintenance and user reassurance.

The operation of this Intelligent Automation System follows a seamless loop of environmental monitoring and automated response, centered on the ESP32 microcontroller. The system is designed to maintain hygiene and comfort in a space—such as a smart restroom—without requiring human intervention.

Detection and Input

The cycle begins with the Sensing Layer. The IR Sensor serves as a proximity trigger, identifying when a user enters or exits the area. Simultaneously, the MQ Gas Sensor performs chemical analysis of the air to detect odors or hazardous gases, while the DHT11 monitors the ambient temperature. These sensors provide a constant stream of data to the ESP32, which acts as the system's "Intelligent Control Unit."

Logic and Execution

The ESP32 processes these inputs against pre-defined thresholds. Its primary role is decision-making. For instance, once the IR sensor confirms a user has departed, the ESP32 calculates the necessary delay before initiating a cleaning cycle. Similarly, if the MQ sensor reports a rise in odor concentration, the ESP32 determines the duration required for ventilation to restore air quality.

Mechanical Response

The final stage is Actuation, where the ESP32's decisions are converted into physical actions. The ESP32 triggers a Relay, which acts as a high-power switch to engage the Pump Motor for an automated flush. Additionally, the Fan is activated to provide ventilation, exhausting odors and regulating temperature based on DHT11 and MQ sensor data. Throughout this process, the LCD Display provides real-time status updates, showing gas levels or system "Ready" states.

In summary, the system transforms raw environmental data into mechanical actions, ensuring a hands-free, hygienic, and well-ventilated environment through efficient electronic control.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

Recent advancements in Internet of Things (IoT), embedded systems, wireless communication, and automation technologies have significantly improved the development of intelligent monitoring systems in modern public infrastructure. Smart monitoring systems are now widely used in healthcare, industrial automation, transportation, environmental monitoring, and smart city applications. Among these areas, public sanitation and hygiene management have become important research topics because hygiene directly affects public health, environmental safety, and passenger comfort.

Railway toilets are one of the most challenging public sanitation environments due to heavy passenger usage, irregular maintenance schedules, poor ventilation, unpleasant odor accumulation, and water wastage. Traditional railway toilet maintenance systems mainly depend on manual cleaning and fixed maintenance intervals instead of actual environmental conditions or usage frequency. This often results in delayed cleaning operations, unhygienic toilet conditions, and poor passenger experience.

To overcome these problems, researchers have started integrating IoT technologies, embedded systems, sensors, and automation mechanisms into smart sanitation systems. These systems use sensors to continuously monitor environmental conditions such as odor levels, temperature, humidity, gas concentration, and user occupancy. The collected data is processed by embedded controllers such as Arduino, Raspberry Pi, and ESP32 to perform intelligent monitoring and automated control operations.

The proposed project, Smart Railway Washroom Automation System is designed based on these modern technological developments. The system integrates sensors, real-time

monitoring, automated flushing, and ventilation control using the ESP32 microcontroller to improve sanitation standards in railway toilets. The following literature survey discusses various existing technologies, research works, methodologies, and limitations related to IoT-based hygiene monitoring and sanitation automation systems.

HYGIENE MONITORING SYSTEMS - IoT-based hygiene monitoring systems have gained significant attention because of their ability to provide real-time monitoring, automation, and intelligent control. Researchers have developed several smart hygiene monitoring systems for public restrooms, hospitals, educational institutions, railway stations, airports, and commercial buildings.

One important research work titled “*An IoT-Based Hygiene Monitoring System in the Restroom*” proposed a sensor-based restroom monitoring framework using gas sensors, humidity sensors, and wireless communication technologies. The system continuously monitored environmental conditions such as odor levels, temperature, and humidity inside restroom facilities. The collected sensor data was processed using embedded controllers and transmitted through wireless communication modules for maintenance monitoring.

The study highlighted the importance of real-time monitoring in improving public hygiene standards. Maintenance staff could identify unhygienic conditions quickly and perform cleaning operations whenever required. The system also demonstrated how IoT technologies can reduce manual inspection and improve operational efficiency.

Another research study focused on smart restroom systems integrated with wireless sensor networks and cloud platforms. The researchers used sensors to monitor restroom occupancy, environmental conditions, and sanitation status. Data collected from multiple restroom units was transmitted to cloud platforms for centralized monitoring and data visualization.

These systems improved monitoring efficiency and allowed authorities to analyze restroom conditions remotely. However, most existing systems mainly focused on monitoring and maintenance notification rather than intelligent automation. Automated flushing systems and ventilation control mechanisms were not fully integrated into many of these solutions.

Several prototype systems also used mobile applications for displaying real-time restroom conditions and maintenance alerts. Such applications improved maintenance coordination and enabled remote supervision. However, internet dependency and implementation cost remained important challenges.

SMART PUBLIC TOILET MONITORING SYSTEMS - Public toilet monitoring systems are important for maintaining hygiene standards in crowded public places such as railway stations, bus terminals, airports, shopping malls, and commercial complexes. Traditional public toilet maintenance methods mainly rely on fixed cleaning schedules and manual inspection. These methods are inefficient because they do not consider actual usage frequency or environmental conditions.

Researchers have proposed several smart public toilet monitoring systems using IoT and embedded technologies to improve sanitation management. A research study titled “*Smart Public Restroom/Toilets Monitoring System*” proposed an intelligent sanitation monitoring framework using PIR motion sensors, MQ gas sensors, and wireless communication technologies.

The system continuously monitored restroom occupancy and odor levels using sensors connected to microcontrollers such as Arduino and ESP32. Whenever environmental conditions exceeded predefined threshold values, the system generated alerts for maintenance staff through GSM and WiFi communication modules. LCD displays were also used for showing environmental status and system information.

Researchers have also developed smart toilet management systems using cloud computing technologies and wireless sensor networks. These systems stored environmental data on cloud platforms and enabled centralized monitoring of multiple sanitation facilities simultaneously. Real-time data visualization and alert generation helped improve maintenance coordination.

Despite these advantages, many public toilet monitoring systems still face important limitations. Most systems mainly focus on monitoring and maintenance alerts rather than automated hygiene control. Intelligent flushing systems, automatic ventilation mechanisms, and predictive maintenance support are often missing in existing solutions.

Another important limitation is scalability. Many prototype systems are designed only for small-scale applications and may not perform efficiently in highly crowded environments such as railway toilets. Railway sanitation systems require continuous environmental monitoring, automated control mechanisms, and efficient water management because of heavy passenger usage.

The proposed system is designed to overcome these limitations by integrating multiple functionalities such as user detection, odour monitoring, automated flushing, and ventilation control into a single embedded solution suitable for railway applications.

IoT-BASED AIR QUALITY MONITORING SYSTEMS - Air quality monitoring systems are among the most common applications of IoT technologies because environmental pollution directly affects human health and comfort. Researchers have developed several low-cost air quality monitoring systems using gas sensors, embedded controllers, wireless communication technologies, and cloud computing platforms.

One important study titled “*Low-Cost IoT System for Dense Air Pollution Monitoring Network*” proposed a sensor-based framework for monitoring harmful gases and

environmental pollutants. The system used MQ gas sensors, temperature sensors, humidity sensors, and wireless communication technologies for collecting environmental data continuously.

The collected sensor data was transmitted to cloud platforms for storage, visualization, and analysis. Researchers used mobile applications and web dashboards for displaying environmental conditions in real time. These systems helped identify pollution levels quickly and supported preventive environmental monitoring strategies.

Several air quality monitoring systems use sensors such as MQ gas sensors, smoke sensors, CO₂ sensors, humidity sensors, and temperature sensors. Embedded controllers such as ESP32, NodeMCU, Arduino, and Raspberry Pi are commonly used for sensor integration and wireless communication.

Researchers have highlighted several advantages of IoT-based air quality monitoring systems. Continuous environmental monitoring improves public safety and helps identify hazardous conditions quickly. Wireless communication and cloud integration also support centralized monitoring and large-scale environmental management.

Although these systems effectively monitor air quality and environmental conditions, they mainly focus on pollution analysis rather than sanitation automation. Most systems do not include automated flushing mechanisms, user occupancy monitoring, or intelligent ventilation systems required for railway hygiene management.

Another challenge observed in many IoT-based monitoring systems is internet dependency. Continuous internet connectivity is required for cloud communication and remote monitoring. In railway environments, unstable network connectivity may affect system performance and reliability.

Power management and maintenance complexity are also important concerns in large-scale IoT deployments. Efficient power optimization techniques are necessary for maintaining long-term system operation.

Despite these limitations, the technologies used in air quality monitoring systems provide an important foundation for developing smart sanitation automation systems. Gas sensors and real-time monitoring techniques used in these systems directly support odor monitoring and ventilation control mechanisms in the proposed TRACKMASTER system.

EMBEDDED SYSTEMS FOR AUTOMATION - Embedded systems are widely used in modern automation technologies because of their reliability, compact size, low cost, and real-time processing capabilities. Embedded systems combine hardware and software components to perform dedicated tasks efficiently. These systems are commonly used in healthcare devices, industrial automation, transportation systems, robotics, smart homes, and public infrastructure management.

Researchers have used embedded controllers such as Arduino, Raspberry Pi, NodeMCU, and ESP32 in various IoT-based monitoring and automation systems. These controllers help collect sensor data, process environmental information, and control automated devices based on predefined conditions.

Among these controllers, ESP32 has become highly popular because of its advanced features and wireless communication capabilities. ESP32 supports built-in WiFi and Bluetooth communication, multiple input-output pins, low power consumption, and high processing speed. It can easily interface with various sensors and actuators.

Several research studies have used ESP32 in applications such as environmental monitoring, industrial automation, healthcare systems, smart homes, and public

sanitation management. The controller is highly suitable for real-time monitoring applications because it can process sensor data quickly and support wireless

SENSOR TECHNOLOGIES USED IN SANITATION SYSTEMS - Sensor technologies play an important role in modern sanitation automation systems. Sensors help detect environmental conditions and user activities inside public sanitation facilities. These sensors continuously monitor odour levels, user occupancy, temperature conditions, humidity, and air quality.

PIR motion sensors are commonly used for occupancy detection and user monitoring applications. These sensors detect infrared radiation emitted by human bodies and help determine restroom usage frequency. PIR sensors are widely used because of their low power consumption, low cost, and simple operation.

MQ gas sensors are widely used for detecting harmful gases, smoke, and unpleasant odor levels. These sensors are highly sensitive and can easily interface with embedded controllers such as ESP32 and Arduino. In smart sanitation systems, MQ sensors help identify odor concentration levels and automatically activate ventilation systems when required.

DHT11 sensors are used for measuring temperature and humidity conditions. These sensors provide digital output and require low power consumption. Temperature and humidity monitoring help maintain comfortable restroom conditions and improve environmental analysis. Researchers have also proposed advanced sensor integration methods using ammonia sensors, water level sensors, air quality sensors, and ultrasonic sensors for improving monitoring accuracy and automation efficiency.

The integration of multiple sensors improves real-time monitoring and intelligent decision-making in smart sanitation systems. The proposed TRACKMASTER system

integrates PIR, MQ gas, and DHT11 sensors for real-time hygiene monitoring and automation.

AUTOMATED FLUSHING SYSTEMS - Automated flushing systems are important components of modern sanitation management because they improve hygiene conditions and reduce manual operation. Traditional flushing systems depend on direct user interaction, which may result in poor hygiene and unnecessary water wastage.

Researchers have proposed several sensor-based automated flushing systems using motion sensors, infrared sensors, and embedded controllers. These systems automatically activate the flushing mechanism whenever toilet usage is detected.

Most systems use PIR or infrared sensors for detecting user presence. After the user leaves the restroom, the flushing system is activated automatically. This touch-free operation improves hygiene standards and reduces the spread of germs.

Researchers have highlighted several advantages of automated flushing systems, including improved cleanliness, reduced manual operation, efficient water management, and enhanced user convenience. Automated systems also ensure consistent sanitation maintenance and reduce maintenance workload.

VENTILATION CONTROL SYSTEMS - Proper ventilation is essential for maintaining air quality and reducing unpleasant odor accumulation in public sanitation facilities. Poor ventilation often creates uncomfortable and unhygienic restroom environments, especially in highly crowded railway toilets.

Researchers have developed automated ventilation systems using gas sensors and embedded controllers. These systems continuously monitor gas concentration levels and activate ventilation fans automatically whenever odor levels exceed predefined threshold values.

MQ gas sensors are commonly used for detecting odor concentration levels inside restroom environments. Embedded controllers such as Arduino and ESP32 process sensor data and control ventilation systems automatically.

Several researchers have integrated wireless communication technologies into ventilation systems for remote monitoring and centralized control. However, many systems still lack full integration with automated flushing and sanitation monitoring mechanisms.

2.2 LIMITATIONS OF SYSTEMS

The literature survey shows that existing sanitation monitoring systems still have several limitations. Most systems focus either on environmental monitoring or sanitation automation individually rather than integrating all functionalities into a single solution.

Many systems lack automated flushing mechanisms and intelligent ventilation control. Some systems only provide monitoring and alert generation without performing automated control operations. High implementation cost, maintenance complexity, and scalability limitations are also common challenges.

Several systems are designed mainly for indoor environmental monitoring and are not optimized for railway sanitation applications. Many prototype systems also lack predictive maintenance support and centralized monitoring capabilities.

CHAPTER 3

METHODOLOGY

3.1 PROPOSED SYSTEMS

The proposed system, “Smart Railway Washroom Automation System” is designed to improve hygiene and sanitation in railway toilets through real-time monitoring and automated control mechanisms. Existing railway toilet maintenance systems mainly depend on manual inspection and fixed cleaning schedules, which are inefficient in handling real-time environmental changes and passenger usage patterns. These systems often result in delayed cleaning, unpleasant odour, excessive water wastage, and poor passenger comfort.

To overcome these limitations, the proposed system integrates IoT sensors, embedded processing, and automation technologies. The system continuously monitors environmental conditions such as odour level, temperature, and user activity using multiple sensors. Based on the collected data, the system automatically controls flushing and ventilation mechanisms to maintain proper hygiene conditions.

The proposed system uses the following major technologies and components:

- ESP32 Microcontroller
- PIR Motion Sensor
- MQ Gas Sensor
- DHT11 Temperature Sensor
- Water Pump for Auto Flush
- Ventilation Fan
- LCD Display Module

The ESP32 acts as the central controller that processes all sensor data and executes rule-based automation logic. The PIR sensor detects user presence and

usage count, while the MQ gas sensor measures odour and harmful gas levels. The DHT11 sensor monitors temperature conditions inside the toilet environment. Based on these sensor readings, the ESP32 automatically activates the water pump for flushing operations and controls the ventilation fan for air circulation and odour removal.

The system is designed to operate independently without requiring continuous internet connectivity, making it suitable for railway environments. The use of sensor-based monitoring also ensures passenger privacy by avoiding camera-based systems.

There are two major stages in the proposed system:

- Environmental Monitoring
- Automated Hygiene Control

Environmental Monitoring - Environmental monitoring is the first stage of the system. In this stage, multiple sensors continuously collect real-time data from the toilet environment. The PIR sensor detects user movement and tracks usage frequency. The MQ gas sensor detects odour concentration and harmful gases, while the DHT11 sensor measures temperature conditions.

The ESP32 processes the sensor readings and evaluates the overall hygiene condition of the toilet. Continuous monitoring helps in identifying unhygienic conditions immediately and enables timely automated responses.

Examples of monitored parameters include:

- User presence and usage count
- Odor and gas levels
- Temperature conditions
- Ventilation requirements

This stage ensures accurate real-time condition analysis for maintaining proper hygiene standards.

Automated Hygiene Control - Automated hygiene control is the second stage of the system. Based on the analyzed sensor data, the ESP32 automatically controls flushing and ventilation systems without manual intervention.

When high odour levels or repeated usage are detected, the system activates the ventilation fan to improve air quality. Similarly, the water pump performs automatic flushing operations after user detection to maintain cleanliness while reducing unnecessary water consumption.

The automation process improves efficiency and ensures better resource utilization. The system also displays live operational status and sensor readings on the LCD module for easy monitoring.

Advantages of automated hygiene control include:

- Reduced manual intervention
- Efficient water management
- Improved passenger hygiene experience
- Real-time automatic response
- Better sanitation maintenance

3.2 HARDWARE REQUIREMENTS

- ESP32 Microcontroller
- PIR Motion Sensor
- MQ Gas Sensor
- DHT11 Temperature Sensor
- Water Pump
- Ventilation Fan
- LCD Display

- Power Supply Unit
- Connecting Wires and Breadboard

3.3 SOFTWARE REQUIREMENTS

- Arduino IDE
- Embedded C Programming
- ESP32 Board Package
- Serial Monitor
- IoT Sensor Libraries

3.4 SYSTEM ARCHITECTURE

1) Sensor Data Collection

The sensors continuously collect environmental data such as user presence, gas levels, and temperature conditions from the toilet environment. The collected data is transmitted to the ESP32 controller for processing.

2) Overall Architecture

The overall architecture of the system consists of sensors, ESP32 controller, automated actuators, and display modules. The ESP32 acts as the central processing unit that receives sensor inputs, analyzes environmental conditions, and activates the flushing and ventilation systems automatically. The LCD display provides real-time monitoring information and system status updates.

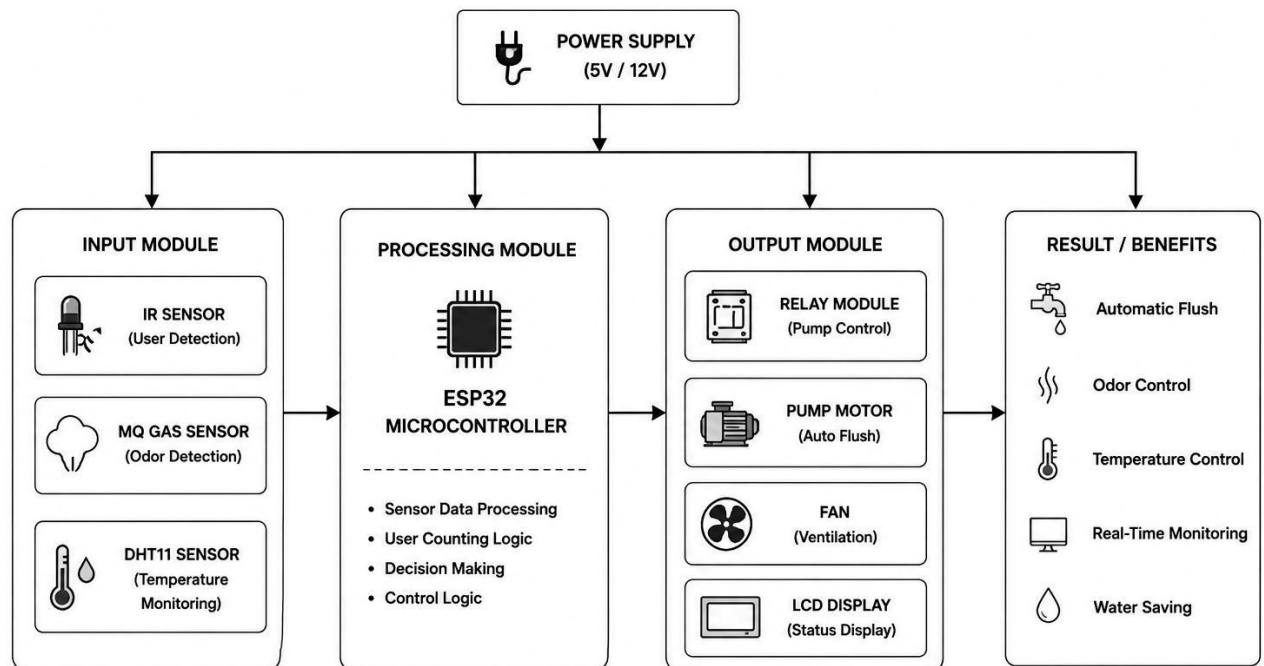


FIG 3.4 OVERALL SYSTEM ARCHITECTURE

The above diagram represents the overall system architecture of the proposed project which explains the interaction between different hardware modules, sensors, processing units, and output devices used for maintaining hygiene and sanitation in railway toilets through automation and real-time monitoring. The system is designed using IoT and embedded system technologies to reduce manual intervention, improve cleanliness, and provide an efficient hygiene management solution.

The architecture is divided into four major sections:

- Input Module
- Processing Module

➤ Output Module

➤ Result / Benefits Module

All these modules are powered using a regulated power supply unit which provides the required operating voltage for sensors, controllers, and output devices.

POWER SUPPLY MODULE

The power supply unit is the primary source of energy for the entire system. It provides regulated DC voltage to all electronic components such as sensors, ESP32 microcontroller, relay module, fan, LCD display, and water pump motor. The diagram shows a **5V / 12V power supply**, where different components operate at different voltage levels according to their requirements.

The ESP32 microcontroller and sensors generally operate at 5V or 3.3V, while the pump motor and ventilation fan may require 12V power for proper functioning. The power supply ensures stable operation of the complete system and prevents fluctuations that may affect sensor readings or automated control operations.

A reliable power supply is important because the system performs continuous real-time monitoring and automation inside railway toilets. Any interruption in power can affect the efficiency of hygiene management and sensor communication.

INPUT MODULE

The input module is responsible for collecting real-time environmental and usage-related data from the railway toilet environment. The system uses multiple

sensors for monitoring hygiene conditions continuously. These sensors act as the primary data collection units of the system.

The input module contains:

- IR Sensor (User Detection)
- MQ Gas Sensor (Odor Detection)
- DHT11 Sensor (Temperature Monitoring)

Each sensor performs a specific monitoring function.

IR SENSOR – USER DETECTION

The IR sensor is used for detecting the presence of users inside the toilet. The sensor works by transmitting and receiving infrared radiation. Whenever a person enters or uses the toilet, the IR sensor detects movement or occupancy and sends signals to the ESP32 controller.

The main purpose of the IR sensor is to monitor toilet usage frequency and activate automation functions when required. The system uses this information to determine whether flushing operations need to be performed automatically.

The IR sensor plays an important role in reducing unnecessary water usage because the flushing mechanism is activated only when actual usage is detected. This improves water management and increases system efficiency.

The sensor also helps maintain hygiene because users do not need to manually operate the flushing system, thereby reducing physical contact with restroom surfaces.

MQ GAS SENSOR – ODOR DETECTION

The MQ gas sensor is used to detect unpleasant odors and harmful gases inside the toilet environment. Railway toilets often experience odor accumulation due to poor ventilation and frequent usage. Continuous odor monitoring is therefore essential for maintaining comfortable and hygienic conditions.

The MQ sensor continuously monitors gas concentration levels in the surrounding environment. When the odor level exceeds a predefined threshold value, the sensor sends signals to the ESP32 microcontroller.

Based on the received sensor values, the controller automatically activates the ventilation fan to improve air circulation and reduce unpleasant smell inside the restroom.

The MQ gas sensor improves passenger comfort and helps maintain better air quality. It also reduces the need for manual odor monitoring by maintenance staff.

DHT11 SENSOR – TEMPERATURE MONITORING

The DHT11 sensor is used for monitoring temperature conditions inside the railway toilet environment. Temperature monitoring is important because high temperatures can create uncomfortable restroom conditions and may increase odor accumulation.

The sensor continuously measures environmental temperature and transmits the readings to the ESP32 controller for processing.

The collected temperature data helps maintain proper environmental conditions inside the restroom. In future implementations, humidity monitoring and

advanced environmental analysis can also be integrated using the same sensor technology.

The DHT11 sensor provides low-cost and reliable environmental monitoring support for the proposed automation system.

PROCESSING MODULE

The processing module is the central controlling unit of the system. In the proposed architecture, the **ESP32 microcontroller** acts as the brain of the complete hygiene automation system.

The ESP32 receives input data from all connected sensors and performs real-time processing based on predefined decision-making logic.

The processing module performs the following operations:

- Sensor Data Processing
- User Counting Logic
- Decision Making
- Control Logic

SENSOR DATA PROCESSING

The ESP32 continuously receives sensor values from the IR sensor, MQ gas sensor, and DHT11 sensor. The controller processes this data in real time and compares the values with predefined threshold levels.

For example:

- If the IR sensor detects user presence, the controller updates the usage count.

- If the MQ sensor detects high odor concentration, the controller activates the ventilation system.
- If environmental conditions change significantly, the controller updates the system status.

This continuous monitoring and processing help maintain proper hygiene conditions automatically.

USER COUNTING LOGIC

The system uses occupancy information from the IR sensor to estimate usage frequency. The ESP32 keeps track of how many times the toilet has been used.

This user-counting logic is important because railway toilets experience different levels of usage throughout the day. Usage-based monitoring helps improve cleaning efficiency and reduces unnecessary flushing operations.

Instead of relying on fixed cleaning schedules, the system performs intelligent automation based on actual usage conditions.

DECISION MAKING

The ESP32 controller performs decision-making operations using predefined control algorithms. The controller determines which output devices need to be activated based on sensor conditions.

Examples include:

- Activating the flushing system after toilet usage.
- Switching on the ventilation fan when odor levels increase.

- Updating system information on the LCD display.

This automated decision-making process improves operational efficiency and reduces the need for manual intervention.

CONTROL LOGIC

The control logic section manages communication between sensors and output devices. The ESP32 sends control signals to the relay module, pump motor, fan, and LCD display according to sensor readings and system conditions.

The controller ensures that all output devices operate only when required, thereby improving water and energy efficiency.

OUTPUT MODULE

The output module contains devices that perform physical operations based on commands from the ESP32 controller.

The output devices include:

- Relay Module
- Pump Motor
- Fan
- LCD Display

RELAY MODULE

The relay module acts as an electrical switching device between the ESP32 controller and high-power devices such as the water pump and fan. Since the ESP32 cannot directly control high-voltage devices, the relay module is used to

safely switch these devices ON and OFF based on control signals from the controller.

The relay ensures proper electrical isolation and protects the microcontroller from voltage damage.

PUMP MOTOR – AUTO FLUSH

The pump motor is responsible for performing the automatic flushing operation. Whenever the controller detects toilet usage through the IR sensor, the flushing mechanism is activated automatically.

This automatic flushing operation improves hygiene and eliminates the need for manual flushing. It also ensures consistent cleaning after each usage.

The system helps reduce water wastage because flushing occurs only when required.

FAN – VENTILATION

The ventilation fan is activated whenever the MQ gas sensor detects high odor concentration levels. The fan improves air circulation inside the railway toilet and helps remove unpleasant odors. Proper ventilation improves passenger comfort and maintains better environmental conditions.

The automatic operation of the fan also improves energy efficiency because the fan operates only when odor levels exceed threshold limits.

LCD DISPLAY – STATUS DISPLAY

The LCD display module is an important output component of the proposed “TRACKMASTER – Smart Train Toilet Hygiene Automation System.” It is used to provide real-time information regarding the operational status of the system

and the current sensor readings. The LCD acts as a user interface between the system and maintenance personnel by displaying important monitoring information continuously.

The display module receives processed data from the ESP32 microcontroller and shows various system parameters in an easy-to-understand format. This helps maintenance staff quickly identify the current condition of the railway toilet without manually checking each component individually.

The LCD display may show the following information:

- User detection status
- Odor level alerts
- Temperature readings
- Flush operation status
- Ventilation fan status
- System active/inactive indication

The LCD module also indicates whether the automatic flushing system and ventilation fan are currently activated. This real-time monitoring feature improves system transparency and helps maintenance personnel observe the complete functioning of the system efficiently.

CHAPTER 4

DESIGN

4.1 UML DIAGRAMS

Unified Modelling Language (UML) diagrams are graphical representations used to visualize, design, and understand the structure and behaviour of a system. UML diagrams help developers and designers represent different functionalities, workflows, and interactions within a system in a clear and organized manner. These diagrams simplify complex system architecture by representing components, processes, and relationships using standardized graphical symbols.

In the proposed project, UML diagrams are used to represent the operational flow, system structure, and interaction between hardware components and automated processes. The UML model helps in understanding how the sensors, ESP32 controller, and actuators communicate with each other to maintain railway toilet hygiene efficiently.

UML diagrams are mainly categorized into two types:

- Structure Diagrams
- Behaviour Diagrams

Structure diagrams represent the static components of the system and their relationships. These diagrams show how different hardware and software modules are organized within the system architecture. Behaviour diagrams represent the dynamic behaviour and workflow of the system during execution. These diagrams describe how data flows through the system and how the system reacts to different environmental conditions.

The UML diagrams used in this project include:

- Structure Chart
- Flow Chart

These diagrams help in understanding the complete working methodology of the proposed smart hygiene automation system.

4.1.1 STRUCTURE CHART

A Structure Chart (SC) is a hierarchical diagram that represents the breakdown of a system into smaller manageable modules. It illustrates the relationship between different functional units of the system and shows how data and control flow between modules. Structure charts are widely used in system design to simplify complex operations and improve system organization.

In the proposed “TRACKMASTER – Smart Train Toilet Hygiene Automation System,” the structure chart represents the interaction between sensors, controller modules, and automated control systems. The ESP32 microcontroller acts as the central processing unit that receives input from multiple sensors and controls the output devices based on sensor readings.

The major modules included in the structure chart are:

- Sensor Module
- ESP32 Processing Module
- Decision and Control Module
- Automated Flush System
- Ventilation Control System
- LCD Display Module

The sensor module collects environmental data such as user presence, odor level, and temperature conditions. The ESP32 controller processes this data using predefined logic and determines the required action. Based on the analysis, the system automatically activates the water pump for flushing and controls the ventilation fan to maintain hygiene conditions. The LCD module displays the real-time operational status of the system.

The structure chart helps in understanding the organization and functional hierarchy of the complete system.

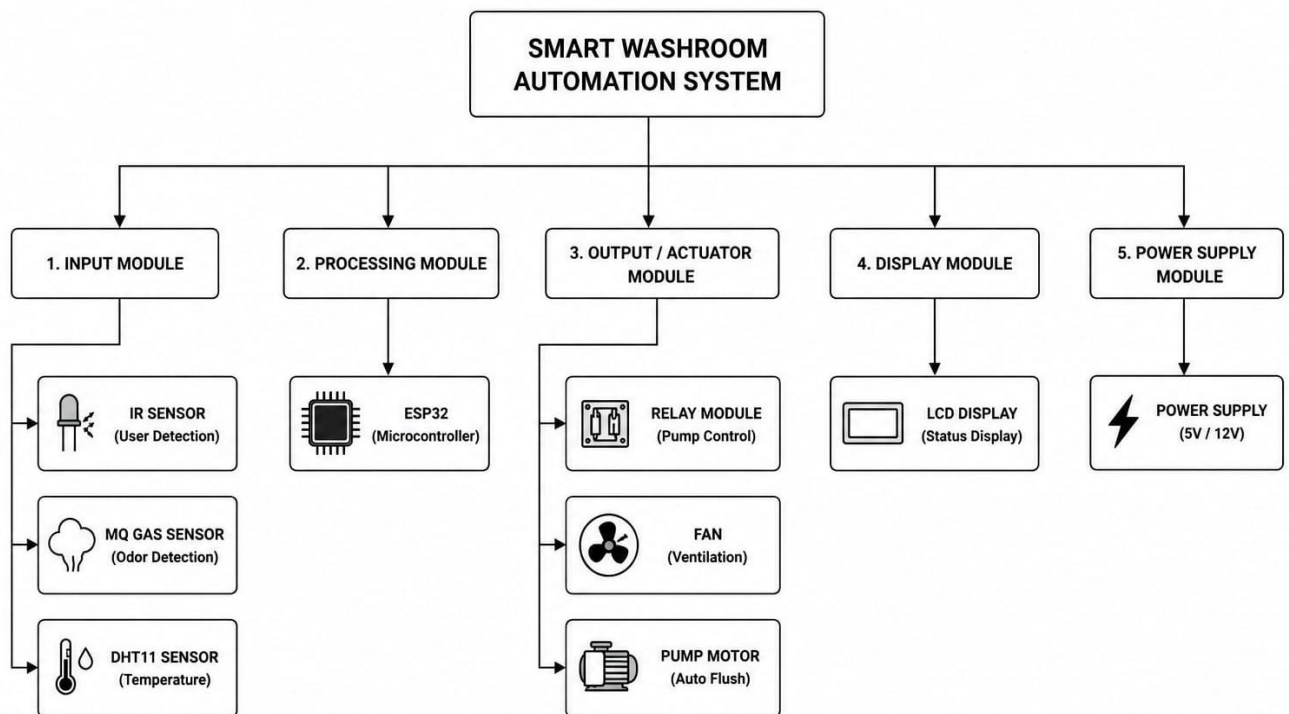


FIG 4.1 STRUCTURE CHART

4.1.2 FLOW CHART

A Flow Chart is a graphical representation of the sequence of operations performed within a system. It represents the workflow of a process using symbols and arrows to show the order of execution. Flowcharts are useful for understanding the logic, decision-making process, and operational flow of a system.

In the proposed project, the flow chart explains the working process of the smart railway toilet hygiene automation system. The process begins with the initialization of sensors and the ESP32 controller. The sensors continuously collect environmental data such as user detection, gas levels, and temperature conditions.

The collected data is processed by the ESP32 microcontroller. The controller checks whether the sensor values exceed predefined threshold levels. If user presence is detected, the system monitors the usage count and activates the automatic flushing mechanism when required. If the gas sensor detects high odour levels, the ventilation fan is automatically switched on to improve air circulation.

The system continuously updates and displays the current status through the LCD module. This process repeats continuously to ensure real-time monitoring and automated hygiene maintenance. The flow chart provides a clear understanding of the sequential operation and automated decision-making process of the proposed system.

FLOW DIAGRAM

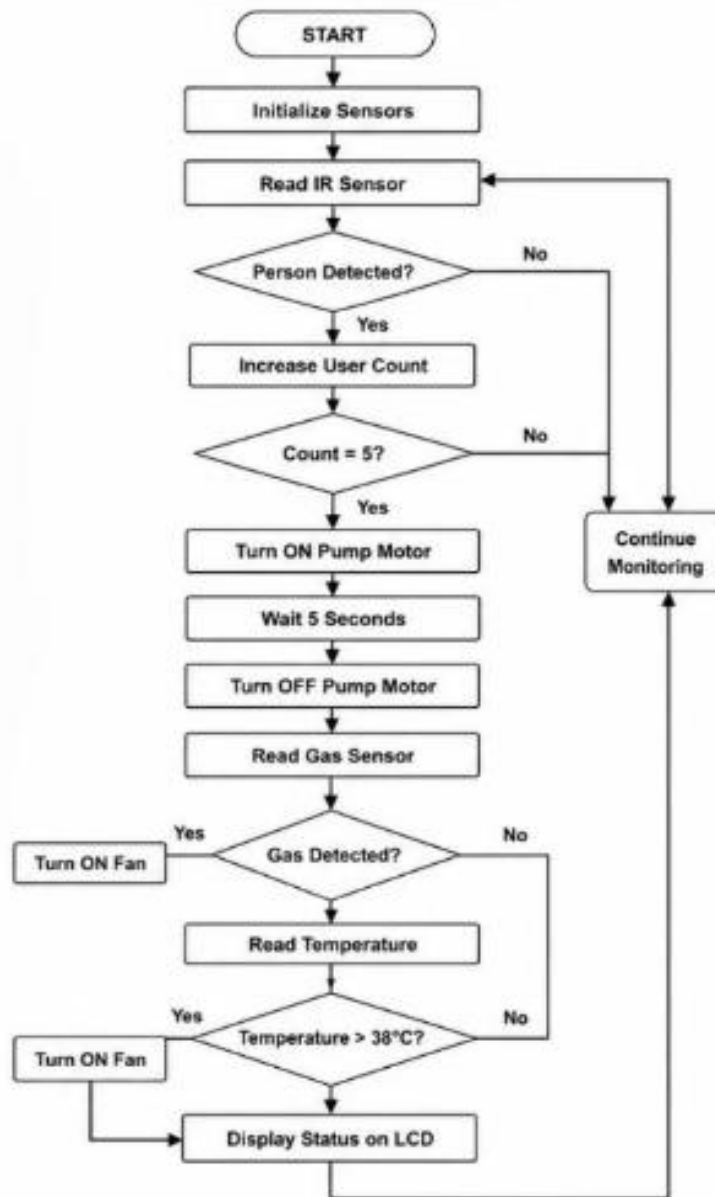


FIG 4.1.2 FLOWCHART OF SMART WASHROOM

CHAPTER 5

EXPERIMENTAL ANALYSIS AND RESULTS

5.1 SYSTEM CONFIGURATION

The proposed project, “Smart Railway Washroom Automation System,” is implemented using embedded hardware components and IoT-based monitoring technologies. The system is designed to operate efficiently on low-power hardware while providing real-time monitoring and automated hygiene control. The ESP32 microcontroller acts as the core processing unit responsible for collecting sensor data, processing environmental conditions, and controlling automated devices such as the flushing pump and ventilation fan.

The system continuously monitors toilet conditions using multiple sensors including PIR, MQ gas sensor, and DHT11 temperature sensor. Based on the sensor readings, the ESP32 processes the data and performs automated operations. The testing and implementation of the system were carried out using embedded programming through Arduino IDE.

The experimental setup consists of real-time sensor monitoring, automatic flushing operation, odour detection, ventilation control, and LCD status display. The system was tested under different environmental conditions and usage patterns to evaluate its performance, response time, and operational efficiency. The project was implemented using a standard computer system for programming and testing purposes. The sensor initialization and program uploading process require only a few minutes, while the real-time execution and automated control operations are performed instantly by the ESP32 controller.

5.2 HARDWARE REQUIREMENTS

The hardware components required for implementing the proposed system are listed below:

- ESP32 Microcontroller
- PIR Motion Sensor
- MQ Gas Sensor
- DHT11 Temperature Sensor
- Water Pump Module
- Ventilation Fan
- LCD Display
- Breadboard and Connecting Wires
- Power Supply Unit
- Personal Computer or Laptop for Programming

System Specifications

- RAM: 4 GB or above
- Storage: 500 GB
- Processor: Intel i5 or higher
- CPU Speed: 2 GHz or faster
- Architecture: 32-bit or 64-bit

5.3 SOFTWARE REQUIREMENTS

The software tools used for the implementation and testing of the proposed system are as follows:

- Arduino IDE
- Embedded C Programming
- ESP32 Board Package
- Serial Monitor
- Sensor Libraries for ESP32
- Operating System: Windows 7 and above / Linux / macOS

The Arduino IDE is used for writing, compiling, and uploading the embedded program to the ESP32 microcontroller. Sensor libraries are integrated to support communication between the ESP32 and connected hardware components.

LIBRARIES AND SOFTWARE TOOLS

The software libraries used in the proposed system help in sensor communication, data processing, and display operations. These libraries simplify the integration of hardware components with the ESP32 controller.

The major libraries used are:

- Wifi Library
- LiquidCrystal_I2C Library
- DHT Sensor Library
- ESP32 Board Libraries
- Wire Library

WIFI LIBRARY

The Wifi library enables wireless communication capabilities in the ESP32 microcontroller. It supports IoT-based connectivity and allows future integration with cloud platforms and remote monitoring systems.

LIQUIDCRYSTAL_I2C LIBRARY

The LiquidCrystal_I2C library is used for interfacing the LCD display module with the ESP32 controller. It enables the display of real-time sensor values, operational status, and system alerts.

DHT SENSOR LIBRARY

The DHT sensor library is used for reading temperature data from the DHT11 sensor. It simplifies communication between the sensor and the ESP32 microcontroller and provides accurate environmental monitoring.

ESP32 BOARD LIBRARIES

The ESP32 board libraries provide support for programming and controlling the ESP32 microcontroller within the Arduino IDE environment. These libraries enable GPIO control, Wifi functionality, and sensor integration.

WIRE LIBRARY

The Wire library is used for I2C communication between the ESP32 controller and peripheral devices such as the LCD display module.

5.4 SOURCE CODE IMPLEMENTATION

The source code of the proposed system is written using Embedded C in the Arduino IDE platform. The program continuously reads sensor data from the PIR sensor, MQ gas sensor, and DHT11 temperature sensor. Based on predefined threshold conditions, the ESP32 controller activates the flushing pump and ventilation fan automatically.

The source code performs the following operations:

- Sensor Initialization
- Real-Time Data Collection
- User Detection
- Odor Detection

- Temperature Monitoring
- Automated Flush Control
- Ventilation Fan Control
- LCD Status Display

The program operates continuously to ensure uninterrupted hygiene monitoring and automatic maintenance operations.

SOURCE CODE

```
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

#define DHTPIN 4
#define DHTTYPE DHT11
#define GAS_SENSOR 34
#define IR_SENSOR 27
#define RELAY_PIN 26
#define FAN_PIN 18

DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27,16,2);

int irCount = 0;
bool pumpRunning = false;

void setup()
```

```

{
    Serial.begin(115200);

    pinMode(GAS_SENSOR, INPUT);
    pinMode(IR_SENSOR, INPUT);
    pinMode(RELAY_PIN, OUTPUT);
    pinMode(FAN_PIN, OUTPUT);
    dht.begin();

    lcd.init();
    lcd.backlight();
    lcd.print("SMART WASHROOM");
}

void loop()
{
    float temperature = dht.readTemperature();
    int gasValue = digitalRead(GAS_SENSOR);
    int irValue = digitalRead(IR_SENSOR);

    if(gasValue == LOW || temperature > 38)
    {
        digitalWrite(FAN_PIN, HIGH);
    }
    else

```

```

{
    digitalWrite(FAN_PIN, LOW);
}

if(irValue == LOW)
{
    irCount++;
}

if(irCount % 5 == 0)
{
    digitalWrite(RELAY_PIN, LOW);
    delay(5000);
    digitalWrite(RELAY_PIN, HIGH);
}

lcd.setCursor(0,0);
lcd.print("Temp:");
lcd.print(temperature);

delay(300);
}

```


5.5 INPUT AND OUTPUT

The input to the proposed system consists of environmental data collected from multiple sensors installed inside the railway toilet. The PIR sensor detects user movement and usage count, the MQ gas sensor measures odour levels, and the DHT11 sensor monitors temperature conditions.

The ESP32 controller processes these inputs and generates corresponding outputs automatically. Based on the sensor readings, the system activates the flushing mechanism and ventilation fan whenever necessary.

Input Parameters

- User Presence Detection
- Odor Level Detection
- Temperature Monitoring

Output Operations

- Automatic Flush Activation
- Ventilation Fan Activation
- LCD Status Display
- Real-Time Environmental Monitoring

The experimental results showed that the system successfully detected environmental changes and responded automatically within a short time. The automated operations improved hygiene maintenance while reducing manual intervention and water wastage.

The experimental results demonstrate that the proposed system can successfully perform real-time environmental monitoring, intelligent decision-making, and automated hygiene control operations with high efficiency and reliability. The integrated sensors continuously collect data related to user presence, odor levels, and

temperature conditions, while the ESP32 microcontroller processes this information accurately and performs the required control actions automatically. The system effectively activates the flushing mechanism and ventilation fan whenever necessary, ensuring proper cleanliness and improved air quality inside the railway toilet environment.

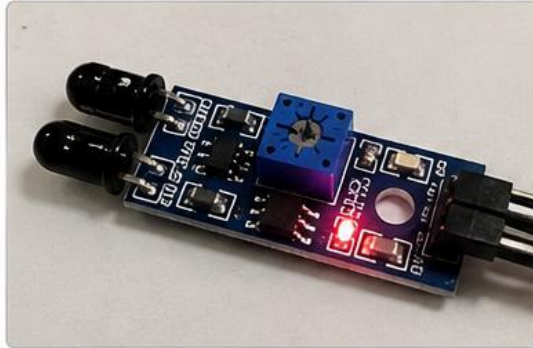
The integration of PIR sensors, MQ gas sensors, DHT11 sensors, automated flushing systems, ventilation control units, and LCD monitoring significantly improves sanitation standards and operational efficiency. The automated functionality reduces dependency on manual inspection and maintenance while also minimizing unnecessary water wastage and energy consumption. Real-time monitoring further helps maintenance staff identify hygiene conditions quickly and ensures faster response to sanitation-related issues.

The system also provides a more comfortable and hygienic experience for railway passengers by maintaining cleaner restroom conditions and reducing unpleasant odor accumulation. Since the proposed solution is based on low-cost IoT and embedded technologies, it can be implemented economically in railway coaches and other public sanitation facilities.

Overall, the successful implementation of the input and output modules confirms that the proposed is capable of providing an intelligent, efficient, low-cost, and scalable smart hygiene management solution for railway toilets and future public sanitation infrastructure. The project demonstrates the practical application of automation and IoT technologies in improving modern sanitation systems and supporting the development of smarter public transportation environments.

INPUTS

Sensors used to detect the environment and user



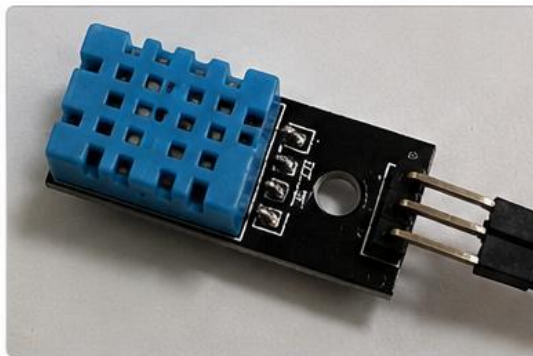
IR SENSOR (User Detection)

Detects the presence of a person entering the washroom.



MQ GAS SENSOR (Odor Detection)

Detects harmful gases / bad odor in the washroom.



DHT11 SENSOR (Temperature)

Measures the temperature inside the washroom.

OUTPUTS

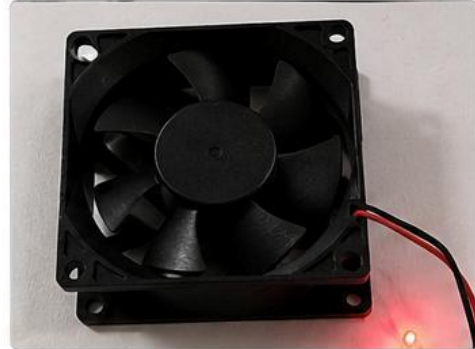
Actions taken by the system based on inputs

LCD DISPLAY



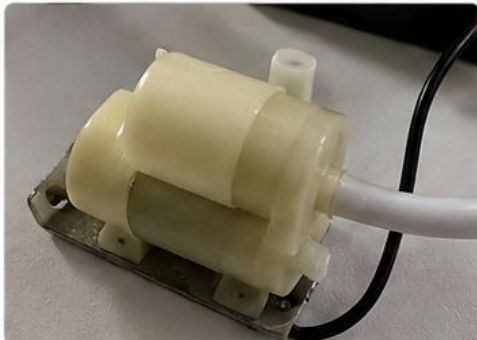
Shows real-time status and system information.

FAN (VENTILATION)



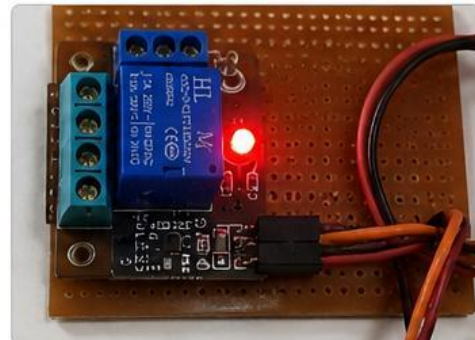
Turns ON when gas is detected or temperature is too high.

PUMP MOTOR (AUTO FLUSH)



Turns ON automatically every 5 users (5 seconds).

RELAY MODULE



Controls the pump motor based on ESP32 commands.

FIG 5.5.1 INPUT AND OUTPUT RESULTS

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The proposed “Smart Railway Washroom Automation System” provides an intelligent and automated solution for improving sanitation and hygiene standards in railway toilets using IoT and embedded technologies. Traditional railway toilet maintenance systems mainly rely on fixed cleaning schedules and manual inspection methods, which often result in delayed cleaning, unpleasant odour, water wastage, and poor passenger comfort. To overcome these limitations, the proposed system integrates real-time environmental monitoring and automated hygiene control mechanisms.

The system uses an ESP32 microcontroller along with PIR, MQ gas, and DHT11 sensors to monitor user presence, odour levels, and temperature conditions continuously. Based on the sensor readings, the system automatically activates the flushing mechanism and ventilation fan whenever necessary to maintain cleanliness and proper air quality. The LCD display module provides real-time monitoring information and operational status updates.

The implementation and testing of the system demonstrated reliable performance and effective automated response capabilities. The proposed system reduces manual intervention, improves water management, enhances passenger comfort, and ensures better hygiene maintenance. Overall, the project successfully combines IoT, automation, and embedded technologies into a cost-effective and scalable solution for smart railway sanitation management and modern public hygiene systems.

6.2 FUTURE WORK

The proposed “Smart Railway Washroom Automation System” can be further enhanced using advanced technologies and intelligent automation techniques. Although the system effectively performs real-time monitoring and automated hygiene control, future developments can improve system efficiency, scalability, and intelligence for large-scale railway sanitation applications.

Artificial Intelligence and Machine Learning algorithms can be integrated to predict cleaning requirements based on passenger usage patterns and environmental conditions. This predictive approach can help railway authorities optimize maintenance schedules and improve operational efficiency. Mobile application integration can also be added to provide remote monitoring, real-time alerts, and maintenance notifications for railway staff.

Future versions of the system can include cloud-based data storage and centralized monitoring platforms for managing sanitation systems across multiple trains and stations. Additional sensors such as humidity sensors, water level sensors, and advanced air quality sensors can improve environmental analysis accuracy. Automatic self-cleaning systems and disinfectant spray mechanisms may also be implemented to reduce manual maintenance further.

With continuous advancements in IoT, automation, and embedded technologies, the proposed system can become a more intelligent, adaptive, and fully automated solution for smart public sanitation and railway infrastructure management.

APPENDIX I

SOURCE CODE FOR ESP32:

```
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

#define DHTPIN 4
#define DHTTYPE DHT11

#define GAS_SENSOR 34    // Now used as DIGITAL input
#define IR_SENSOR 27

#define RELAY_PIN 26
#define FAN_PIN 18

DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);

float temperature = 0;
int gasValue = 0;
int irValue = 0;

bool pumpRunning = false;
unsigned long pumpStartTime = 0;
```

```

int irCount = 0;          // Person counter
bool lastIRState = HIGH; // For edge detection

void setup()
{
    Serial.begin(115200);

    pinMode(GAS_SENSOR, INPUT); // Digital now
    pinMode(IR_SENSOR, INPUT);

    pinMode(RELAY_PIN, OUTPUT);
    pinMode(FAN_PIN, OUTPUT);

    digitalWrite(RELAY_PIN, HIGH);
    digitalWrite(FAN_PIN, LOW);
    dht.begin();
    lcd.init();
    lcd.backlight();
    lcd.setCursor(0, 0);
    lcd.print("SMART WASHROOM");
    lcd.setCursor(0, 1);
    lcd.print("SYSTEM START");

    delay(3000);

```



```

    lcd.clear();
}

void loop()
{
    temperature = dht.readTemperature();

    gasValue = digitalRead(GAS_SENSOR); // Digital read
    irValue = digitalRead(IR_SENSOR);

    bool gasDetected = (gasValue == LOW); // LOW = gas detected
    (typical modules)
    bool heatDetected = (temperature > 38);

    // ----- FAN CONTROL -----
    if (gasDetected || heatDetected)
        digitalWrite(FAN_PIN, HIGH);
    else
        digitalWrite(FAN_PIN, LOW);

    // ----- IR PERSON COUNT (Edge Detection) -----
    if (irValue == LOW && lastIRState == HIGH)
    {
        irCount++;
        Serial.print("Person Count: ");
    }
}

```

```

Serial.println(irCount);

if (irCount % 5 == 0 && !pumpRunning)
{
    pumpRunning = true;
    pumpStartTime = millis();
    digitalWrite(RELAY_PIN, LOW);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("COUNT: ");
    lcd.print(irCount);
    lcd.setCursor(0, 1);
    lcd.print("PUMP ON");
}
}

lastIRState = irValue;

// ----- PUMP TIMER -----
if (pumpRunning && (millis() - pumpStartTime >= 5000))
{
    digitalWrite(RELAY_PIN, HIGH);
    pumpRunning = false;

    lcd.clear();

```

```

    lcd.setCursor(0, 0);
    lcd.print("PUMP OFF");
    delay(1000);
}

// ----- LCD STATUS -----

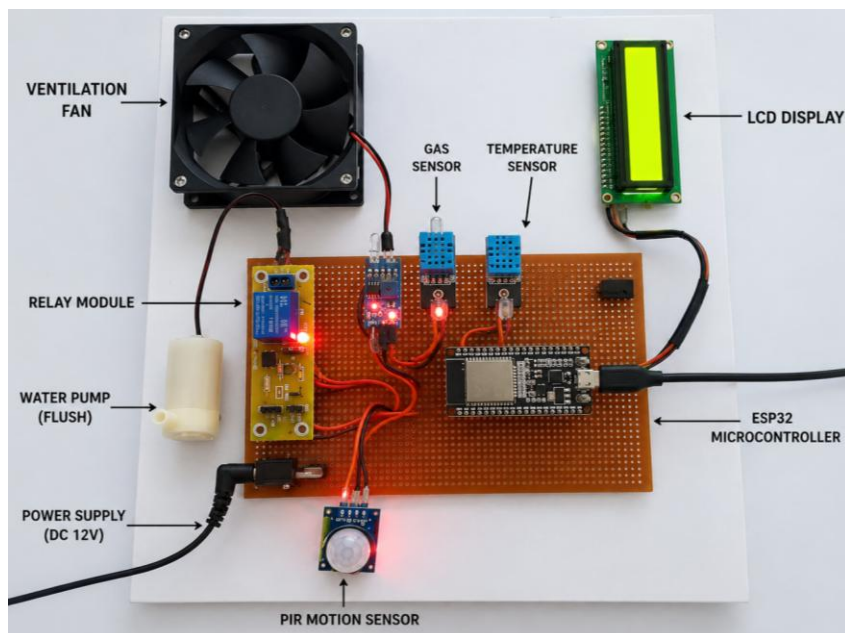
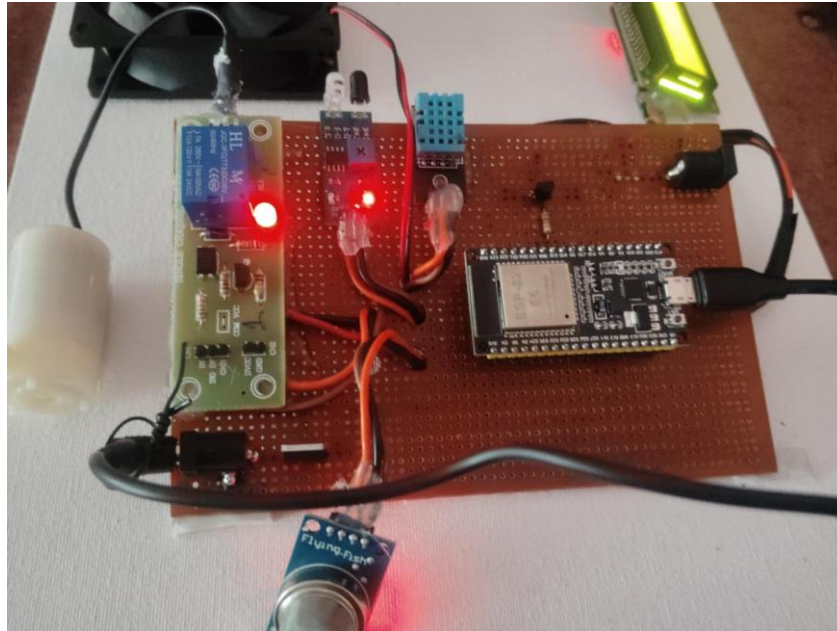
lcd.setCursor(0, 0);
lcd.print("T:");
lcd.print(temperature);
lcd.print((char)223);
lcd.print("C ");
lcd.print("C:");
lcd.print(irCount);
lcd.setCursor(0, 1);
if (gasDetected)
    lcd.print("SMELL ALERT ");
else if (heatDetected)
    lcd.print("HIGH TEMP ");
else
    lcd.print("NORMAL ");

delay(300);
}

```

APPENDIX II

OUTPUT SNIPPETS:



REFERENCES

- [1] Ahmed Khan.,et al. (2023). “Restroom Hygiene Smart Monitoring System Incorporating LoRa Technology.” ResearchGate Publications.
- [2] Anjali P.,et al. (2024). “E-Toilet Using IoT.” International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE).
- [3] Arun K.,et al. (2024). “Design of IoT-Based Smart Bio-Toilet Scheme.” Journal of Information and Communication Technology.
- [4] Harish Kumar.,et al. (2025). “IoT Based Autonomous Solution for Maintenance of Public Toilets.” SCITEPRESS Conference Proceedings.
- [5] Joseph Mathew.,et al. (2024). “Intelligent Toilet System for Enhanced Hygiene and Comfort Using IoT.” ResearchGate Publications.
- [6] Priya Nair.,et al. (2020). “Smart Toilet Monitoring System using IoT.” International Journal of Computer Trends and Technology (IJCTT).
- [7] Rajesh Kumar.,et al. (2024). “An IoT-Based Hygiene Monitoring System in the Restroom.” ResearchGate Publications.
- [8] Rakesh S.,et al. (2024). “Smart Public Toilet Management and Monitoring System using IoT.” ResearchGate Publications.
- [9] Sharma, R.P.,et al. (2024). “Smart Public Restroom/Toilets Monitoring System.” International Journal of Advanced Research in Science, Communication and Technology (IJARSCT).
- [10] Vivek Raj.,et al. (2024). “Smart Toilet Monitoring System using IoT for Hygiene and Health.” Journal of Smart Technology and Management Journals.